Unconscious Movement

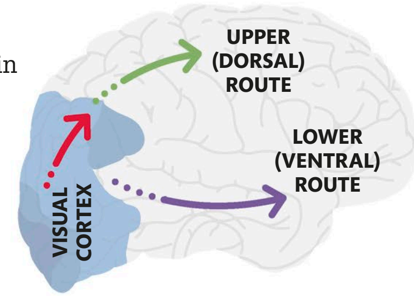
We perform many voluntary actions without having to think about them because they are so familiar. Another kind of unconscious movement is the reflex action—an instinctive response to danger.

WHY DOES BEING TIRED SLOW DOWN OUR REACTION TIME?

When we are tired, neurons in our brain slow down, affecting our visual perception and memory. This means we respond to events more slowly.

Reaction pathways

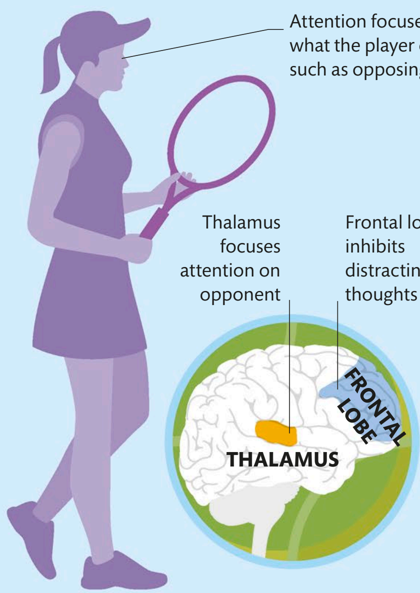
Visual information is vital in helping us plan our movements. Information from the visual cortex follows two routes in the brain (see pp.70–71). The upper (or dorsal) route, which leads to the parietal lobe, guides our actions in real time. Meanwhile, the lower (or ventral) route, which ends at the temporal lobe, triggers stored visual experiences to help us interpret what we see and respond accordingly.



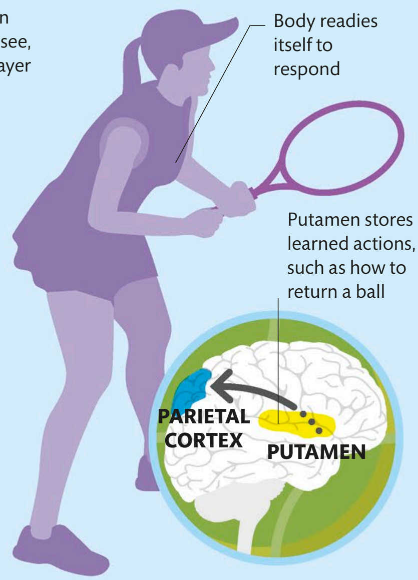
Visual pathways in the brain
The dorsal route carries information on the position of the body and other objects, while the ventral route draws on perception and memory for identifying objects. The brain uses this information to judge the strength and direction required for a movement.

Coordinated actions

Any sequence of actions demands coordination between different parts of the brain—first to focus attention on the task, then to integrate sensory information and memory to create a plan, then to engage the motor area to act. Acquiring a new skill, such as driving or playing a sport, involves learning and practicing movement sequences so that they become almost unconscious. When we learn a skill, our brain cells form new connections. By the time we have mastered a skill (see box, right), there is far less cortical activity associated with performing that task than there was when we were a novice. As a result, the actions of a skilled person—such as a professional tennis player—are more rapid, precise, and subtle.



1 Attention
To prepare for action, the thalamus directs attention to the area where the activity will occur (such as the opposing player), while the frontal lobes block distracting thoughts so the player can concentrate on the visual cues.



2 Memory
Visual cues trigger the parietal cortex to call up memories of action sequences from the putamen. The parietal cortex uses this information to assess the context and create an internal model for the action.